Sensor-Independent Hyperspectral Target Detection With Semisupervised Domain Adaptive Few-Shot Learning

Yanzi Shi, *Graduate Student Member, IEEE*, Jiaojiao Li, *Member, IEEE*, Yunsong Li, *Member, IEEE*, and Qian Du, *Fellow, IEEE*

***Abstract*— Deep learning-based hyperspectral target detection (HTD) is potentially hindered by the limited training samples and sensor-dependent transferability. To address this issue, we propose a novel semisupervised domain adaptive few-shot learning (SDAFL) model to adaptively transfer similarity/dissimilarity measurement from source domain with sufficient labeled samples to target domain in an adversarial manner, where source data and target data can be collected by different sensors, i.e., sensor-independent. In order to alleviate negative transfer, residual channel attention (RCA) and weighted domain adaptation (WDA) are used to automatically select representative features and assign easy-transferred samples with higher priority. In addition, we adopt modulated deformable convolution (MDConv) to make the receptive field fit image spatial structure and also introduce a discriminatively boosted loss (DBL) function based on the prior known target signature to further enhance feature distinction, where intraclass similarity is improved, while interclass similarity is suppressed. After extracting discriminative features through the SDAFL model, guided filter and t-distribution kernel are jointly used for spatial–spectral target detection (S^2TD). It should be noted that only the spectral signature of the desired object is needed in the target domain. Experimental results and analysis on three real hyperspectral images (HSIs) verify the efficiency and superiority of our proposed sensor-independent hyperspectral target detection (SIHTD) method compared with other algorithms.**

***Index Terms*— Discriminatively boosted loss (DBL), modulated deformable convolution (MDConv), semisupervised domain adaptive few shot, sensor-independent hyperspectral target detection (SIHTD).**

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Yanzi Shi, Jiaojiao Li, and Yunsong Li are with the State Key Laboratory of Integrated Service Network, Xidian University, Xi'an 710071, China (e-mail: yzshi_xidian@163.com; jjli@xidian.edu.cn; ysli@mail.xidian.edu.cn).

Qian Du is with the Department of Electrical and Computer Engineering, Mississippi State University, Starkville, MS 39762 USA (e-mail: du@ece.msstate.edu).

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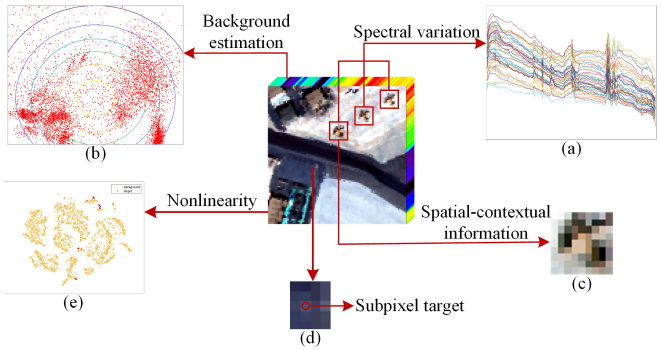


Fig. 1. Illustration of five main challenges in HTD. (a) Spectral signatures of the same type of airplanes remain different. (b) Pixels of the first two principal components are represented as red circle, and the corresponding MND contours are plot in parula color. (c) Shape, size, and texture information of the desired target. (d) Mixture of neighboring background and target in subpixel. (e) Background and target pixels are plotted in orange and red circles, respectively. They are linearly inseparable.

I. Introduction

HYPERSPECTRAL target detection (HTD) has been demonstrated its importance in the fields of vegetation and geologic survey, civilian search and rescue, environmental risk assessment, and urban planning [1]–[4]. Different from the box-/patch-level object detection, HTD aims to make soft decisions for each pixel based on abundant spectral or joint spatial–spectral information of a specific target, which mainly refers to pixel-/subpixel-level detection. There are several critical challenges involved in the HTD technique as shown in Fig. 1, mainly including: 1) spectral variation, namely, spectral signatures collected from the same object may be different due to atmospheric compensation, geometric distortion, sensor noise, and other factors [5]; 2) background distribution/covariance estimation with limited pure samples, particularly in statistical methods [6]; 3) spatial–contextual information, which is usually underestimated in the detection of targets with more than one pixel [3]; 4) subpixel-level object detection, especially in image scenes captured by spaceborne sensors with a relatively low spatial resolution [6]; and 5) nonlinearity exploitation, required further investigation besides kernel learning theory [3].

Numerous approaches have been proposed to address the aforementioned issues and have yielded remarkable detection performance. For the spectral variation issue, several researchers applied multiple target signatures to tolerate more

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spectral deviation, such as iteratively reweighted unstructured detector [7], multitask learning-based reliable detection (MultiRely) [8], and the improved versions of constrained energy minimization (CEM), i.e., multiple-target CEM (MTCEM), sum CEM (SCEM), and winner-take-all CEM (WTACEM) [9]–[11]. Apart from the simple multiple targets strategy, the subspace projection algorithms can decrease intraclass dissimilarity through certain constraints in the embedded feature space [12], [13]. For background characterization problem, classical statistical methods are based on multivariate normal distribution (MND), such as adaptive matched filter (AMF) [14] and adaptive coherence estimator (ACE) [15]. However, such an assumption is generally invalid in practice according to Fig. 1(b). Therefore, many other background models have been proposed, e.g., rectangularly contoured model, finite mixture model, and subspace model. Several effective methods have emerged based on these background modeling strategies, including single-band anomaly detection (SBAD) [16], student's t mixture model (StMM) [17], and orthogonal subspace projection (OSP) [18]. Recently, with the development of spectroscopic imaging technology, especially airborne, handheld, and truck-mounted spectrometers, taking advantage of spatial and spectral information for pixel-level object detection has attracted more attention. These methods can typically fuse spatial and spectral information in feature level [19], [20] or decision level [21]. However, current spaceborne hyperspectral sensors still have a relatively coarse spatial resolution (usually between 30 and 100 m) for large-scale earth observation (EO), which would lead to many subpixel targets. To address this issue, several methods based on the popular linear/nonlinear mixture model (LMM/NLMM) have been presented [22], [23]. Finally, for the nonlinear structure of hyperspectral data [e.g., background and target pixels are linearly inseparable in the original space referring to Fig. 1(e)], kernel learning has been introduced to extract nonlinear property of hyperspectral image (HSI) and has achieved superior performance to the linear versions [6].

Deep learning has already been successfully applied in a wide range of applications in HSI processing due to its strong nonlinear feature representation ability [24], [25], such as classification [26], superresolution [27], band selection [28], and detection [29], and it has brought a significant breakthrough over many traditional approaches. Compared with other HSI processing tasks, data-driven deep learning-based HTD faces two critical problems: insufficiency of labeled samples and class imbalance where the proportion of the target class is very small in contrast to that of the background class. Hence, direct application of existing deep neural networks to HTD is infeasible. To mitigate this issue, one intuitive idea is data augmentation [29]–[31], which takes difference (similarity and dissimilarity) between paired pixels as training samples. However, the aforementioned work can obtain satisfactory detection accuracy only when both training and testing data captured from the same sensor. Once transferred to different sensors, the performance of such sensor-dependent methods may drop dramatically. On the other hand, unsupervised learning, which only uses unlabeled data to extract representative features beneficial for detection task, has provided valuable insights on deep learning-based HTD [20], [32]. However, these unsupervised models are thoroughly dependent on unlabeled data itself. It is expected that the detection performance can be further improved if similarity/dissimilarity measurement learned from another previously labeled HSI data (e.g., Indian Pines, Pavia University, and Salinas) is used to guide the detection of unlabeled HSIs containing the novel targets.

Recently, few-shot learning has attracted considerable attention for providing a solution to the problem of insufficient labeled training data [33]. The objective of few-shot learning is to solve a specific target task through knowledge learned from abundant labeled source data and a few labeled/unlabeled target data. Existing few-shot learning approaches can be roughly categorized into three groups: 1) model-based methods; 2) metric learning-based methods; and 3) optimization-based methods. Model-based methods fine-tune the parameters trained from the source domain based on a few labeled target samples [34]. Metric learning-based methods mainly concentrate on learning an appropriate feature transformation and then measure the similarity between the support embeddings of known categories and the test embedding through a specific distance metric. Several well-known models have been proposed, including matching network (MatchingNet) [35], prototypical network (ProtoNet) [36], and relation network (RelationNet) [37]. The third optimization-based few-shot learning approaches intend to learn some other novel optimization algorithms instead of traditional gradient descent due to its unfitness in few-shot regime [38]. Nowadays, considerable efforts toward few-shot learning have been done on classifier design with few labeled data. Inspired by this idea, we apply few-shot learning to handle limited labeled and imbalanced training samples in deep learning-based HTD. However, due to the multimodal distributions of data collected by different sensors, a model pretrained in source domain can hardly be applicable for a completely new target domain, i.e., HTD based on few-shot learning is still sensor-dependent. To conquer this issue, researchers developed a domain adaptation technique to achieve global data distribution alignment in the feature embedding space [39] so that the novel domain adaptive few-shot learning (DAFL) framework can transfer similarity/dissimilarity measurement across different domains. Domain adaptation methods can be divided into three categories according to the amount of available labeled target data: 1) supervised: it requires labeled target examples [40]; 2) semisupervised: it leverages from both labeled and available unlabeled target data [41]; and 3) unsupervised: it does not require any labeled samples from target domain [42]. Hence, the integration of few-shot learning and domain adaptation is of great significance for sensor-independent HTD.

In this article, we aim to deal with labeled data insufficiency, class imbalance, and sensor-dependent transferability in deep learning-based pixel-level HTD. Specifically, we propose a novel sensor-independent hyperspectral target detection (SIHTD) method based on the semisupervised DAFL (SDAFL) model, which can exploit the nonlinear spatial–spectral structure of HSI and mitigate spectral variations. The proposed method consists of three steps. First, spatial–spectral similarity/dissimilarity measurement in

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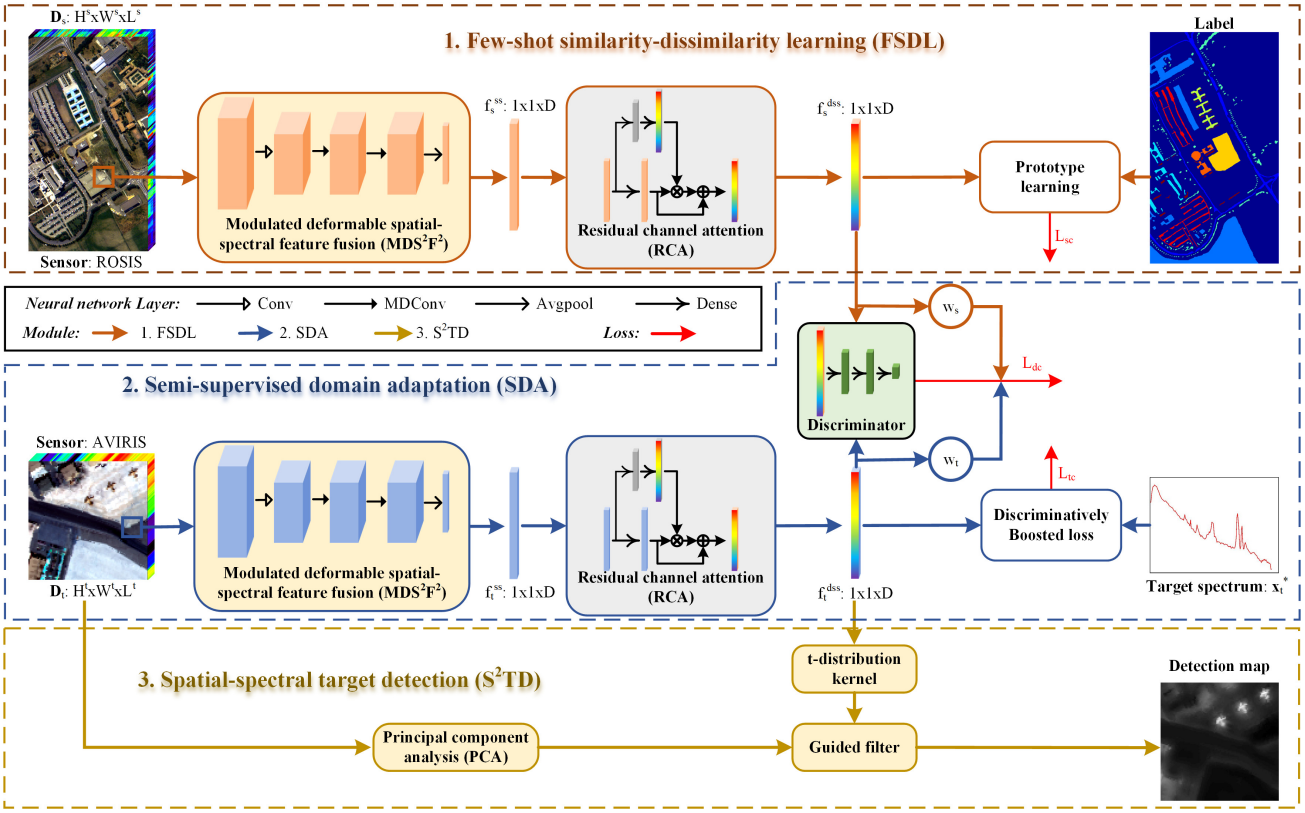


Fig. 2. Overall flowchart of the proposed method based on SDAFL.

embedding space is obtained from source data with abundant labeled training data. Here, we adopt ProtoNet for learning an initial embedding space due to its simplicity and competitiveness. Furthermore, modulated deformable convolution (MDConv) [43] and residual channel attention (RCA) are introduced to improve the ability of discriminative feature representation. Then, in the embedding space, a semisupervised domain adaptation (SDA) module, including weighted domain adaptation (WDA) and discriminatively boosted feature extraction, is designed for global distribution alignment and target-background distinction. Finally, we combine a simple t-distribution kernel detector and a guided filter to further enhance the spatial structure of detection map. The main contributions of this article can be summarized as follows.

1) A new tailored SDAFL model is proposed to exploit the transferability of similarity/dissimilarity measurement for detecting pixel-level target of interest. In particular, no assumptions on distribution of labeled source data and unlabeled target data are required. Consequently, the proposed method is sensor-independent. To the best of our knowledge, this is the first time that DAFL has been exploited for sensor-independent HTD task.
2) To address the key issue in the knowledge transferability between different sensors, RCA mechanism and WDA strategy are proposed to emphasize easy-transferred samples to promote positive transfer.
3) We introduce MDConv and discriminatively boosted loss (DBL) to further improve the discriminative capacity of deep spatial–spectral features for subsequent target detection.

The proposed SIHTD is compared with several state-of-the-art algorithms on three real hypespectral benchmarks. The experimental results and further analysis demonstrate the effectiveness and superiority of SIHTD with only one positive sample.

The remainder of this article is organized as follows. Section II gives a detailed description of the proposed SIHTD method. In Section III, experimental results and analysis on three real HSIs are presented. Finally, the conclusions of this article are drawn in Section IV.

II. Proposed Method

A. Problem Description

Given data from two domains source HSI data $\mathbf{D}_s \in \mathbb{R}^{H^s \times W^s \times L^s}$ with C^s classes and target HSI data $\mathbf{D}_t \in \mathbb{R}^{H^t \times W^t \times L^t}$ with a prior-known signature of interest $\mathbf{x}_t^*$. The distribution of source data $P_s(\mathbf{x}_s)$ and target data $P_t(\mathbf{x}_t)$ can be either the same or different, and $\mathbf{x}_s$ and $\mathbf{x}_t$ denote source and target sample, respectively. In this article, we focus on a more practical case of sensor-independent HTD, i.e., $P_s(\mathbf{x}_s) \neq P_t(\mathbf{x}_t)$. Formally, $\mathbf{D}_s = \{(\mathbf{x}_s^i, y_s^i) | \mathbf{x}_s^i \sim P_s(\mathbf{x}_s), y_s^i \in C^s\}_{i=1}^{H^s \times W^s}$ and $\mathbf{D}_t = \{\mathbf{x}_t^i | \mathbf{x}_t^i \sim P_t(\mathbf{x}_t)\}_{i=1}^{H^t \times W^t}$, where y_s^i denotes the label value of sample $\mathbf{x}_s^i$. The objective is to exploit labeled $\mathbf{D}_s$, unlabeled $\mathbf{D}_t$, and prior-known target spectrum $\mathbf{x}_t^*$ for extracting discriminative features that can perform well in the final target detection task. Note that only N^s labeled source samples per class are randomly selected as training data ($N^s \times C^s \ll H^s \times W^s$). Fig. 2 shows the overall flowchart of the proposed SIHTD method. There are three modules

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designed for HTD, including few-shot similarity–dissimilarity learning (FSDL), SDA, and spatial–spectral target detection (S^2TD), which are discussed in detail in the following.

B. Few-Shot Similarity-Dissimilarity Learning

ProtoNet is adopted as the basic model for FSDL in our SIHTD method due to its simplicity and competitiveness. This neural network transforms the original source HSI into a new feature embedding space through a nonlinear function $\zeta_{\theta_s}^s : \mathbb{R}^{w_p \times w_p \times L^s} \to \mathbb{R}^{D\times 1}$ with learnable parameters θ_s to enhance intraclass similarity and interclass dissimilarity, where w_p is the spatial size of HSI patch. There are two random sampling operations to form a training episode, i.e., class subset selection from C^s classes and support set S_s and query set Q_s selection from N^s examples for each class. In this article, we select 2 classes, 1 support sample per class, and 15 query samples per class (2-way-1-shot) in each episode. Specifically, S_s and Q_s are first fed into ProtoNet to extract high-level features. Then, the prototype of the kth class $\mathbf{c}_s^k \in \mathbb{R}^{D\times 1}$ is calculated by averaging those embedded support features from the same class

$$\mathbf{c}_s^k = \frac{1}{|S_s^k|} \sum_{\left(\mathbf{x}_s^{s_i}, y_s^{s_i}\right)\in S_s^k} \zeta_{\theta_s}^s\left(\mathbf{x}_s^{s_i}\right) \tag{1}$$

where $S_s^k = \{(\mathbf{x}_s^{s_i}, y_s^{s_i}) : (\mathbf{x}_s^{s_i}, y_s^{s_i}) \in S_s, y_s^{s_i} = k\}$ represents the support subset belonging to the kth class, $|S_s^k| = 1$ in this article, and $\mathbf{x}_s^{s_i}$ is the support sample in source domain.

Based on $\zeta_{\theta_s}^s(\cdot)$ and $\mathbf{c}_s^k$, a class distribution for a query sample in source domain $\mathbf{x}_s^{q_i}$ is generated by a softmax function over the distance between the embedded query feature and the prototypes

$$p_{\theta_s}\left(y_s^{q_i} = k \middle| \mathbf{x}_s^{q_i}\right) = \frac{\exp\left(-d\left(\zeta_{\theta_s}^s\left(\mathbf{x}_s^{q_i}\right), \mathbf{c}_s^k\right)\right)}{\sum_{k'} \exp\left(-d\left(\zeta_{\theta_s}^s\left(\mathbf{x}_s^{q_i}\right), \mathbf{c}_s^{k'}\right)\right)} \tag{2}$$

where $d(\cdot)$ denotes the Euclidean distance function and $y_s^{q_i}$ is the predicted class of $\mathbf{x}_s^{q_i}$. The loss function of ProtoNet L_{sc} is the negative log probability of $p_{\theta_s}(y_s^{q_i} = k|\mathbf{x}_s^{q_i})$ defined as

$$L_{sc} = -\sum_{\left(\mathbf{x}_s^{q_i}, y_s^{q_i}\right)\in Q_s} \log p_{\theta_s}\left(y_s^{q_i} = k \middle| \mathbf{x}_s^{q_i}\right). \tag{3}$$

Minimizing (3) can encourage the network to better measure similarity/dissimilarity. However, considering that the HSI cubic data also contain useful spatial information besides abundant spectral information, we apply MDConv whose receptive field can adapt to the complex spatial structure to extract effective spatial–spectral features. MDConv learns adjustable offsets and modulation scalars on the input feature map, which can adaptively manipulate the locations and amplitudes of grid sampling of the standard convolution (Conv), as shown in Fig. 3. Therefore, the network is given the ability to suppress useless spatial information, which may impose additional interferences to feature extraction, e.g., neighboring pixels dissimilar to the center pixel.

Given an input feature map, $\mathbf{x} \in \mathbb{R}^{C_{in}\times H\times W}$, where C_{in}, H, and W denote the channel, height, and width, respectively.

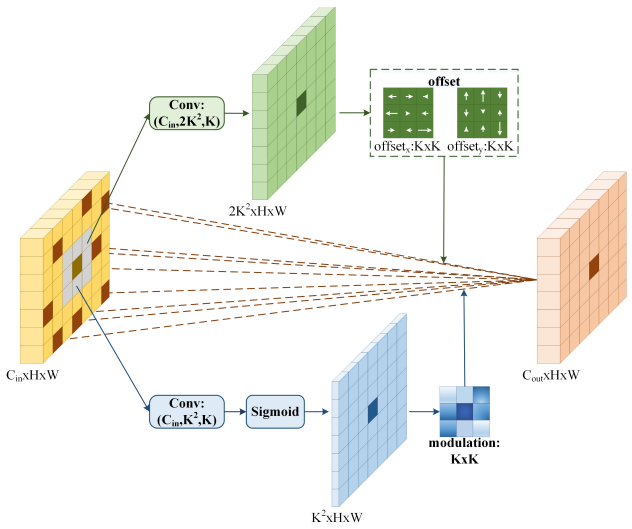


Fig. 3. Illustration of a $K \times K$ MDConv.

The standard Conv with $K \times K$ kernel is formulated as

$$\mathbf{y}(p_x, p_y) = \sum_{i=-K/2}^{K/2} \sum_{j=-K/2}^{K/2} \mathbf{x}(p_x + i, p_y + j) \cdot \mathbf{w}_{ij} \tag{4}$$

where $\mathbf{y}(p_x, p_y)$ denotes the output feature map at location (p_x, p_y) and $\mathbf{w}_{ij}$ is the corresponding kernel weight. It is obvious that the spatial sampling locations are fixed, which cannot model geometric transformations very well.

To address this issue, MDConv manipulates the sampling grid by a pair of learnable offsets $(\Delta i, \Delta j)$ and further reweights each sampling grid with a modulation scalar $\Delta\mathbf{m}_{ij} \in [0, 1]$. The output feature map becomes

$$\mathbf{y}(p_x, p_y) = \sum_{i=-K/2}^{K/2} \sum_{j=-K/2}^{K/2} \mathbf{x}(p_x + i + \Delta i, p_y + j + \Delta j) \cdot \mathbf{w}_{ij} \cdot \Delta\mathbf{m}_{ij} \tag{5}$$

where both $(\Delta i, \Delta j)$ and $\Delta\mathbf{m}_{ij}$ are obtained via standard Convs with $K \times K$ kernel on the input feature map $\mathbf{x}$, and the output channel of the two convolutional layers is $2K^2$ and K^2, respectively. In addition, a sigmoid function is applied to make $\Delta\mathbf{m}_{ij}$ lie in $[0, 1]$. Since the obtained offsets Δi and Δj are generally fractional, bilinear interpolation is exploited to compute $\mathbf{x}(p_x + i + \Delta i, p_y + j + \Delta j)$ as in [44].

For pixel-level HTD task, we design a simple but efficient neural network based on MDConv for spatial–spectral feature fusion (called MDS^{2}F^2), as shown in Fig. 2. The subnetwork architecture consists of three convolutional (one Conv and two MDConvs) layers and one average pooling (Avgpool) layer. In order to accelerate convergence and avoid dead nodes during training, each convolution is followed by batch normalization (BN) and leaky rectified linear unit (LReLU). Taking an HSI patch $\mathbf{x}_s \in \mathbb{R}^{w_p\times w_p\times L^s}$ as input, the output of MDS^{2}F^2 can be represented as

$$\mathbf{f}_s^{ss} = \zeta_{\theta_{s_1}}^s(\mathbf{x}_s) \tag{6}$$

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where $\zeta^s_{\theta_{s_1}}(\cdot)$ denotes the nonlinear transformation function of MDS^{2}F^2 parameterized by θ_{s_1} and $\mathbf{f}_s^{\text{ss}} \in \mathbb{R}^{D\times 1}$ is the extracted D-dimensional spatial–spectral feature.

Furthermore, to highlight easy-transferred features, attention mechanism is adopted in this article. However, directly performing dot products between feature map and attention mask ranging from 0 to 1 may degrade feature representation ability as layers going deeper [45]. Once the attention mask fails to select useful features, the following network can hardly make an effective modification. To alleviate this issue, we present an RCA subnetwork, as shown in Fig. 2. The subnetwork consists of two branches: one mask branch with two dense layers followed by an LReLU and sigmoid activation function, and one trunk branch with one dense layer followed by an LReLU function. In addition to automatically assign different importance to features such as typical channel attention, the identity connection in our RCA unit can also avoid irreversible false suppression on important features and offer the whole network a second opportunity to modify this failure. Combining MDS^{2}F^2 and RCA subnetwork together, the final output of few-shot learning module can be expressed as

$$\mathbf{f}_s^{\text{dss}} = \left(\mathbf{1} + \zeta^s_{\theta_{s_{21}}}\left(\mathbf{f}_s^{\text{ss}}\right)\right) * \zeta^s_{\theta_{s_{22}}}\left(\mathbf{f}_s^{\text{ss}}\right) \tag{7}$$

where $\mathbf{f}_s^{\text{dss}} \in \mathbb{R}^{D\times 1}$, $*$ denotes the dot product, and $\zeta^s_{\theta_{s_{21}}}(\cdot)$ and $\zeta^s_{\theta_{s_{22}}}(\cdot)$ represent the top mask branch and bottom trunk branch, respectively.

In this way, the attention mask is designed as identical mapping similar to residual learning. Thus, the entire FSDL module would not perform worse than its counterparts without attention. It should be noted that the image scene content of source data has little impacts on detection performance since the FSDL module learns to measure similarity/dissimilarity between pixel pairs instead of simply mapping one pixel into a hidden space, i.e., we pay more attention to relative difference between different samples rather than the samples themselves.

C. Semisupervised Domain Adaptation

Once MDS^{2}F^2 and RCA are trained, an SDA module is presented to make the embedding space learned on the source domain more adaptable to differently distributed target data. First, initialize the target few-shot learning module (also including MDS^{2}F^2 and RCA subnetwork) $\zeta^t_{\theta_t} : \mathbb{R}^{w\times w\times L^s} \to \mathbb{R}^{D\times 1}$ with the pretrained source one $\zeta^s_{\theta_s}$, where a robust band selection algorithm named optimal clustering framework (OCF) [46] is applied on target data to ensure the same spectral dimension as the source data. Then, fix the source FSDL module and update $\zeta^t_{\theta_t}$ by applying the SDA technique, which consists of two parts: WDA and discriminatively boosted feature extraction. Details are given in the following.

1) Weighted Domain Adaptation: For further alignment between source distribution $P_s(\mathbf{x}_s)$ and target distribution $P_t(\mathbf{x}_t)$, a powerful discriminator subnetwork ζ_{θ_d} with learnable parameter θ_d is designed, as shown in Fig. 2. It consists of three dense layers followed by the LReLU function (sigmoid for the last layer). The domain adversarial loss function is defined as

$$\begin{aligned} L_{dc} = \min_{\theta_d}\max_{\theta_t} \; & -\mathbb{E}_{\mathbf{x}_s\sim P_s(\mathbf{x}_s)} \log \zeta_{\theta_d}\left(\mathbf{f}_s^{\text{dss}}\right) \\ & -\mathbb{E}_{\mathbf{x}_t\sim P_t(\mathbf{x}_t)} \log \zeta_{\theta_d}\left(1-\zeta^t_{\theta_t}(\mathbf{x}_t)\right) \end{aligned} \tag{8}$$

The purpose of the discriminator ζ_{θ_d} is to distinguish whether the embedded spatial–spectral feature comes from source domain or target domain, while the target FSDL module $\zeta^t_{\theta_t}$ aims to learn domain-confused feature representation. However, the performance of the SDA module would be hindered by simply using (8) since it assigns all samples with the same weight and ignores the transferability of different samples. Motivated by [47], we modify (8) to selectively highlight easy-transferred examples and suppress difficult ones by imposing entropy conditioning. The improved loss function is formulated as

$$\begin{aligned} L_{wdc} = \min_{\theta_d}\max_{\theta_t} \; & -\mathbb{E}_{\mathbf{x}_s\sim P_s(\mathbf{x}_s)} \mathbf{w}(\mathbf{g}_s) \log \zeta_{\theta_d}\left(\mathbf{f}_s^{\text{dss}}\right) \\ & -\mathbb{E}_{\mathbf{x}_t\sim P_t(\mathbf{x}_t)} \mathbf{w}(\mathbf{g}_t) \log \zeta_{\theta_d}\left(1-\zeta^t_{\theta_t}(\mathbf{x}_t)\right) \end{aligned} \tag{9}$$

where $\mathbf{w}(\mathbf{u}) = 1 + e^{-H(\mathbf{u})}$, $H(\mathbf{u}) = \sum_{k=1}^{2} \mathbf{u}_k \log \mathbf{u}_k$ denotes the entropy criterion function, and $\mathbf{u} = \{\mathbf{g}_s, \mathbf{g}_t\}$. Here, $\mathbf{g}_s$ and $\mathbf{g}_t$ are the class probability vectors of an embedded feature in source domain and target domain, respectively, which can be obtained based on Euclidean distance between each source (target) embedded feature and prototypes $\mathbf{c}_s^k$ (prior-known feature of interest $\zeta^t_{\theta_t}(\mathbf{x}_t^*)$)

$$\begin{aligned} \mathbf{g}_s &= \text{Softmax}\left(\left\|\mathbf{f}_s^{\text{dss}} - \mathbf{c}_s^k\right\|_2^2\right) \\ \mathbf{g}_t &= \frac{1}{1+\left\|\mathbf{f}_t^{\text{dss}} - \zeta^t_{\theta_t}\left(\mathbf{x}_t^*\right)\right\|_2^2}. \end{aligned} \tag{10}$$

It can be seen that the easier a sample transfers, the smaller $H(\mathbf{u})$ is, and the higher $\mathbf{w}(\mathbf{u})$ is. Therefore, the WDA technique can make these easy-transferred examples priority to difficult ones and alleviate negative transfer to some extent.

2) Discriminatively Boosted Feature Extraction: Apart from applying WDA to match data distribution as much as possible, discriminative structure of the original target HSI should also be preserved for subsequent target detection. To achieve this goal, a DBL function is designed to further optimize the target FSDL module. First, we generate several background spectra through subspace projection in the original data space. It can be implemented as follows.

Background Spectra Generation Process:

1) *Initialize:* Set $\mathbf{E}_0 = \mathbf{x}_t^*$, $e = 1$. N_{bkg} denotes the number of background samples to be generated.
2) *Background Generation:* At the eth iteration, construct an orthogonal complement subspace through $\mathbf{P}^{\perp}_{\mathbf{E}_{e-1}} = \mathbf{I} - \mathbf{E}_{e-1}(\mathbf{E}_{e-1}{}^T\mathbf{E}_{e-1})^{-1}\mathbf{E}_{e-1}{}^T$, and find a sample with the largest orthogonal projection into subspace $\mathbf{E}_{e-1}$ as background $\mathbf{x}_t^{b_e}$: $\mathbf{x}_t^{b_e} = \arg\{\max_{\mathbf{x}_t} \|\mathbf{P}^{\perp}_{\mathbf{E}_{e-1}}\mathbf{x}_t\|_2^2\}$.
3) *Update:* If $e < N_{bkg}$, let $\mathbf{E}_e = [\mathbf{E}_{e-1}\ \mathbf{x}_t^{b_e}]$, $e = e+1$, and go to step 2; otherwise, let $\mathbf{x}_t^b = [\mathbf{x}_t^{b_1}\ \mathbf{x}_t^{b_2}\ \ldots\ \mathbf{x}_t^{b_{N_{bkg}}}]$ and continue.
4) *Possible Target Removal:* If $((\mathbf{x}_t^{*T}\mathbf{x}_t^{b_e})/(\|\mathbf{x}_t^*\|\|\mathbf{x}_t^{b_e}\|)) > \tau$, remove $\mathbf{x}_t^{b_e}$ from $\mathbf{x}_t^b$.

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TABLE I
SIMPLE EXAMPLE OF DBL

Sample	$i=1$		$i=2$	
Class	$j=1$	$j=2$	$j=1$	$j=2$
$\mathbf{d}_{ij}^{(0)}$	0.5000	0.5000	0.1000	0.9000
$\mathbf{d}_{ij}^{(1)}$	0.5000	0.5000	0.0122	0.9878
$\mathbf{d}_{ij}^{(2)}$	0.5000	0.5000	0.0002→**0**	0.9998→**1**

After obtaining background spectra, we adopt the t-distribution kernel to measure detectability of a sample and further boost the discrimination of the difficult pixels by designing a novel distribution. Specifically, based on the t-distribution kernel, the detectability of a sample can be obtained as

$$\mathbf{d}_{ij} = \frac{1}{1+\left\|\mathbf{f}_t^{\mathrm{dss}_i} - \zeta_{\theta_t}^t\left(\mathbf{x}_t^{qj}\right)\right\|_2^2}$$

and

$$\sum_j \mathbf{d}_{ij} = 1 \tag{11}$$

where $\mathbf{f}_t^{\mathrm{dss}_i}$ denotes the ith feature, $\mathbf{x}_t^q = [\mathbf{x}_t^*\ \mathbf{x}_t^b]$ represents the N_k known target and background signatures in total, $j = 1, 2, \ldots, N_k$. The closer $\mathbf{d}_{i1}$ is to zero or one, and the easier the ith sample can be detected.

In order to transform difficult samples to easier ones, we define a new distribution $\mathbf{h}_{ij}$ from $\mathbf{d}_{ij}$ as follows:

$$\mathbf{h}_{ij} = \frac{\mathbf{d}_{ij}^{\lambda}}{\sum_j \mathbf{d}_{ij}^{\lambda}}. \tag{12}$$

If the original distribution $\mathbf{d}_{ij}$ can converge to the constructed $\mathbf{h}_{ij}$ during training, i.e., $\mathbf{d}_{ij} \to \mathbf{h}_{ij}$, then

$$\lim_{t\to\infty} \mathbf{d}_{ij}^{(t)} = \begin{cases} \dfrac{1}{N_k}, & \text{if } \mathbf{d}_{ij}^{(0)} = \dfrac{1}{N_k}\ \forall j \\ 1, & \text{if } \mathbf{d}_{ij}^{(0)} > \mathbf{d}_{il}^{(0)}\ \forall j \neq l \\ 0, & \text{if } \mathbf{d}_{ij}^{(0)} < \mathbf{d}_{il}^{(0)}\ \forall j \neq l \end{cases} \tag{13}$$

where t denotes the training step. For better illustration, a simple example with $N_k = 2$, $\lambda = 2$ is given in Table I. Therefore, we present a DBL function based on Kullback–Leibler (KL) divergence to restrict $\mathbf{d}_{ij}$ and $\mathbf{h}_{ij}$ to be similar

$$L_{tc} = \sum_{i,j} \mathbf{h}_{ij} \log \frac{\mathbf{h}_{ij}}{\mathbf{d}_{ij}}. \tag{14}$$

Fig. 4 shows the shapes of the novel DBL function under different parameter λ when $N_k = 2$. It can be seen that the larger λ can boost the discriminative capacity of HSI. Moreover, it can be observed that the DBL function would have little effects on the most difficult samples with the same probability to be classified into each class. Fortunately, such samples are quite rare in practical HTD task. In this article, λ is empirically set to 10 for all experiments.

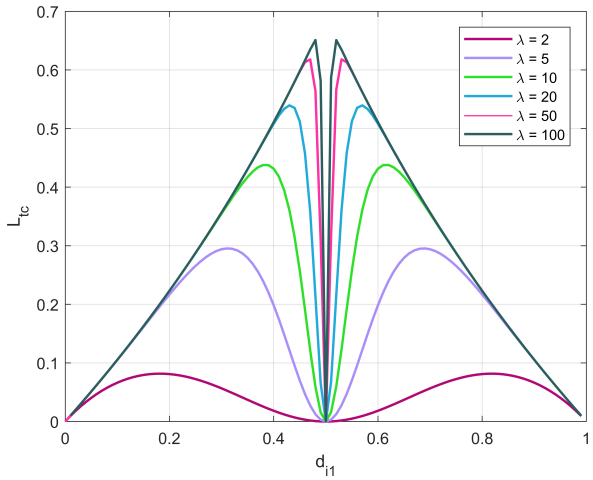


Fig. 4. Shape of the DBL function under different parameters λ.

D. Spatial–Spectral Target Detection

Once the SDA module is well trained, the features learned from the target FSDL module can provide improved discriminative capacity for HTD. In this article, we apply t-distribution kernel to target embeddings for spectral target detection and impose spatial information through the guided filter for complementarity. The detection procedure can be divided into three steps.

1) Discriminative feature generation via the trained target FSDL module

$$\mathbf{f}_t^{\mathrm{dss}_i} = \zeta_{\theta_t}^t\left(\mathbf{x}_t^i\right), \quad i = 1, 2, \ldots, H^t \times W^t \tag{15}$$

where $\mathbf{x}_t^i$ and $\mathbf{f}_t^{\mathrm{dss}_i}$ are the ith original HSI sample and corresponding embedded feature, respectively.

2) Spectral target detection via t-distribution kernel

$$\mathbf{r}_{\mathrm{spe}} = \frac{1}{1+\left\|\mathbf{f}_t^{\mathrm{dss}} - \zeta_{\theta_t}^t\left(\mathbf{x}_t^*\right)\right\|_2^2} \tag{16}$$

where $\mathbf{r}_{\mathrm{spe}}$ is the spectral detection map, $\mathbf{x}_t^*$ denotes the prior-known signature of interest, and $d(\cdot)$ is the Euclidean distance.

3) Spatial structure preservation via guided filter

$$\mathbf{r}_{\mathrm{ss}}^i = \sum_j \mathbf{G}^{ij}(\mathbf{I})\mathbf{r}_{\mathrm{spe}}^j$$
$$\mathbf{G}^{ij}(\mathbf{I}) = \frac{1}{|\omega|^2} \sum_{k:(i,j)\in\omega_k} \left(1 + \frac{(\mathbf{I}_i - \mu_k)(\mathbf{I}_j - \mu_k)}{\sigma_k^2 + \epsilon}\right) \tag{17}$$

where $\mathbf{r}_{\mathrm{ss}}$ is the final spatial–spectral detection map, $\mathbf{G}^{ij}(\mathbf{I})$ denotes the filter kernel, $\mathbf{I}$ is the guidance image setting as the first principal component of the original HSI, ω_k is a window centered at the kth pixel of size $(2r+1)\times(2r+1)$ (r is the radius of the window), ϵ is the regularization parameter, and μ_k, σ_k^2, and $|\omega|$ are the mean and variance of $\mathbf{I}$, and the number of pixels in ω_k, respectively. The impacts of parameters r and ϵ on detection performance are given in Section III-D1.

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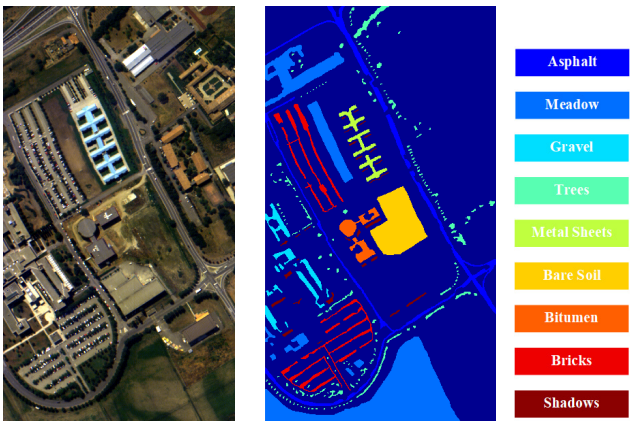


Fig. 5. Source data. (a) Pseudo color image. (b) Ground truth.

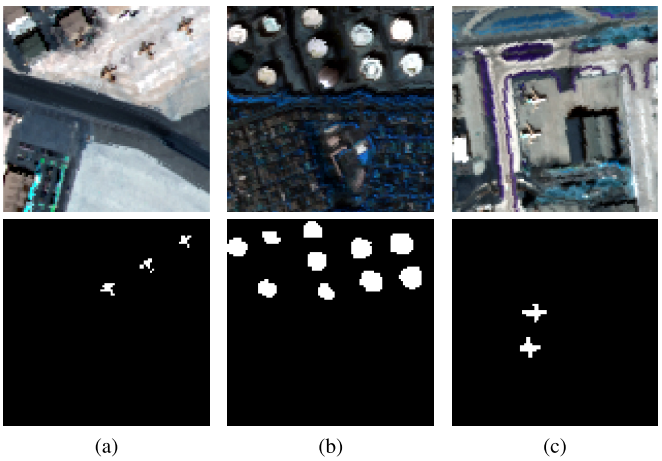


Fig. 6. Target data. (a) AVIRIS-1. (b) AVIRIS-2. (c) AVIRIS-3.

III. Experimental Results

A. Hyperspectral Data

As shown in Section II-A, this article mainly focuses on the case that the source domain and target domain conform to different distributions. Considering that the available HSIs for target detection are mostly captured by the Airborne Visible/InfraRed Imaging Spectrometer (AVIRIS) sensor with the wavelengths of 0.4 μm–2.5 μm, we select hyperspectral data captured by the Reflective Optics System Imaging Spectrometer (ROSIS) ranging from 0.43 to 0.86 μm as source data to evaluate the performance of the proposed SIHTD. The detailed instructions are provided in the following.

1) Source Data: The source data were captured by the ROSIS sensor over the University of Pavia, Pavia, Italy. It consists of 610 × 340 pixels with a spatial resolution of 1.3- and 103-m bands after noisy bands removal. There are totally nine labeled classes, including asphalt, meadow, gravel, trees, metal sheets, bare soil, bitumen, bricks, and shadows in this image scene. Fig. 5 shows the pseudocolor image and the corresponding ground truth.

2) Target Data: Three HSIs collected by the AVIRIS sensor are applied for HTD task. The AVIRIS-1 data set covers the area of San Diego airport, CA, USA, containing 100 × 100 pixels and 189 bands after discarding low SNR and water absorption bands. The spatial resolution is 3.5 m. The second data set AVIRIS-2 was collected over El Segundo, CA, USA. A subimage of size 100 × 100 × 211 with 7.1-m ground sampling distance (GSD) is selected for our experiments. The AVIRIS-3 data set was captured over Los Angeles, CA, USA. This image has 100 × 100 pixels with a spatial resolution of 7.1- and 205-m available bands after eliminating 19 noisy bands. The pseudocolor image and the corresponding ground truth of these three data sets are shown in Fig. 6.

B. Experimental Setup

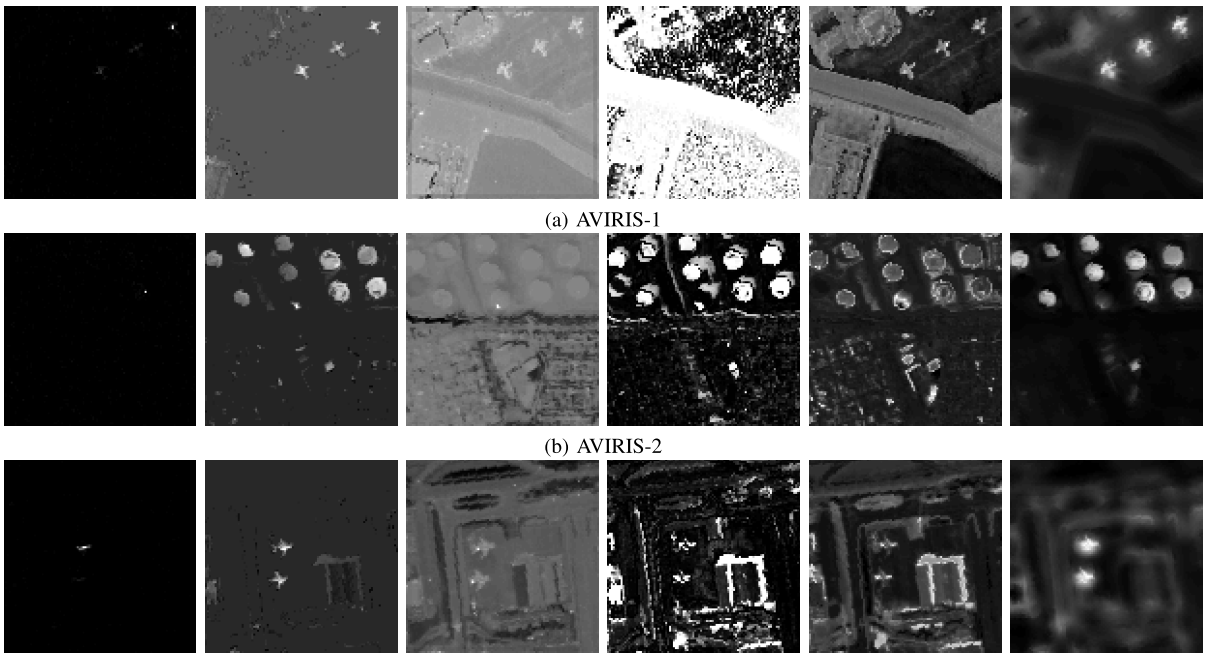
A comparison between the proposed method and five different detection approaches reported in the literature has been conducted for performance evaluation, including: 1) ACE; 2) hierarchical CEM (hCEM)[1] [48]; 3) combined sparse and collaborative representation (CSCR)[2] [49]; 4) convolutional neural network-based detection (CNND)[3] [29]; and 5) deep network-based hyperspectral target detection (HTD-Net)[2] [31]. For hCEM, the parameters λ and ϵ are set as 200 and 10^{-3} in all experiments. In CSCR, the dual-window size $(\omega_{out}, \omega_{in})$ is selected as $(11, 3)$, $(11, 5)$, and $(11, 3)$ for AVIRIS-1, AVIRIS-2, and AVIRIS-3, respectively. For the sake of fairness, the source data in Fig. 5 is used as the reference data in CNND, and the learning rate is chosen from the set $\{0.0005, 0.001, 0.005, 0.01, 0.5\}$. In HTD-Net, the number of the generated similarity and dissimilarity pixel pairs remains the same for each data set, i.e., 31 125 and 32 250, respectively. As for the proposed SIHTD method, we randomly select 180 samples per class in source domain as training data, i.e., $N^s = 180$. The initial learning rate, multistep factor of decay, and weight decay are set to 0.001, 0.2, and 0.0005, respectively. The detailed analysis of other parameters of feature dimension D, patch size w_p, and r and ϵ of guided filter is given in Section III-D. Note that SIHTD is repeatedly implemented ten times for reliable assessment.

In order to quantitatively evaluate the performance of different detectors, the 3-D receiver operating characteristic (3-D ROC) curve [50], [51] and three corresponding 2-D ROC curves (i.e., 2-D ROC curve of (τ, P_F), 2-D ROC curve of (τ, P_D), and 2-D ROC curve of (P_F, P_D)), and the area under curve (AUC) of the three kinds of 2-D ROC curve are used in this article. The variables τ, P_F, and P_D denote threshold, false alarm rate, and detection probability, respectively. The better a detector performs, the higher the AUC of (P_F, P_D) (abbreviated as AUC_1), the higher the AUC of (τ, P_D) (abbreviated as AUC_2), and the lower the AUC of (τ, P_F) (abbreviated as AUC_3). Thus, we further define an overall accuracy named AUC_{oa} [52] to comprehensively measure the detection performance as follows:

$$\text{AUC}_{\text{oa}} = \text{AUC}_1 + \text{AUC}_2 - \text{AUC}_3. \tag{18}$$

[1] http://levir.buaa.edu.cn/code_cn.htm
[2] http://fdss.bit.edu.cn/yjdw/js/153191.htm
[3] https://github.com/yuchaoY/CNND

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(a) AVIRIS-1

(b) AVIRIS-2

(c) AVIRIS-3

Fig. 7. Subjective detection maps of our proposed SIHTD and other competitors on three target data sets. From left to right: ACE, hCEM, CSCR, CNND, HTD-Net, and SIHTD. Note that among the ten repeated SIHTD experiments, the detection map with the closest AUC_1 to the mean AUC_1 is chosen to display here. (a) AVIRIS-1. (b) AVIRIS-2. (c) AVIRIS-3.

Obviously, the higher the AUC_{oa}, the better the detection. In addition, kappa statistic, a measurement of interrater agreement, is exploited for method verification. As HTD usually makes soft decisions instead of hard decisions, we first calculate different kappa coefficient values under different thresholds and then select the maximum kappa for performance evaluation.

With regard to the hardware equipment, a computer consisting of an Intel Xeon E5-1620 v4 processor with 10M of cache and an NVIDIA GeForce GTX 1080Ti GPU is adopted in the experiment. As for the software environment, the detection methods, ACE, hCEM, and CSCR, are implemented on CPU with MatlabR 2017a, while three deep learning-based methods, CNND, HTD-Net, and SIHTD, are carried out in Python 3.6 on GTX 1080Ti GPU.

C. Results and Analysis

1) Qualitative Results: Fig. 7 shows the detection maps obtained by three widely used classical approaches (i.e., ACE, hCEM, and CSCR), two state-of-the-art CNN-based detectors (i.e., CNND and HTD-Net), and our method over three HSIs. As shown in Fig. 7, the detection map indicates that both hCEM and our proposed SIHTD significantly outperform all other approaches. Compared with SIHTD, hCEM seems to slightly outperform SIHTD in background suppression on AVIRIS-1 and AVIRIS-3, but it can hardly deal with the detection of mixed pixels in the edge area, especially on data set AVIRIS-3. More seriously, hCEM faces the critical challenge of completely inverse detection of target and background due to the inaccurate background correlation estimation [48], whereas SIHTD does not need making any assumptions of background. The ACE method omits a great proportion of target pixels mainly due to its inaccurate assumption on background estimation and has a very low tolerance to spectral variation. The deep learning-based CNND and HTD-Net methods fail to suppress false alarm pixels, especially for CNND on data AVIRIS-1. As for CSCR, it cannot provide an intuitive detection result and further process is needed to locate targets since the difference between target and background pixels is relatively small. Considering spatial-contextual information, the proposed SIHTD method deals with target detection in edge areas more effectively than hCEM as shown in Fig. 7(a) and (c) and produces a detection map closer to the ground truth. Hence, the proposed method can obtain satisfactory performance in target preservation and background suppression.

2) Quantitative Evaluation and Analysis: As it is sometimes difficult to decide which method performs better solely on visualization, such as the detection maps of hCEM and SIHTD shown in Fig. 7, Fig. 8 shows four types of ROC curves of all competitors on three data sets as a complementary evaluation. By observing from 2-D ROC curves of (P_F, P_D), we can obtain that SIHTD begins to detect target first, i.e., $P_D > 0$ at the lowest P_F. Furthermore, it achieves 100% detection probability at a lower P_F than all other competitors on three data sets, except for CSCR on AVIRIS-2, which detects all target pixels at a slightly lower false alarm rate. As for 2-D- ROC curves of (τ, P_D) and (τ, P_F), it is not difficult to conclude that our proposed SIHTD performs superior to most comparing methods and achieves better performance than hCEM in all experiments except that in background suppression on AVIRIS-3. However, the consideration of ROC curves alone cannot provide precise evaluation of a detector, as the ROC curves of different algorithms are generally intersectional. For example, the 2-D ROC curve of (τ, P_D) of

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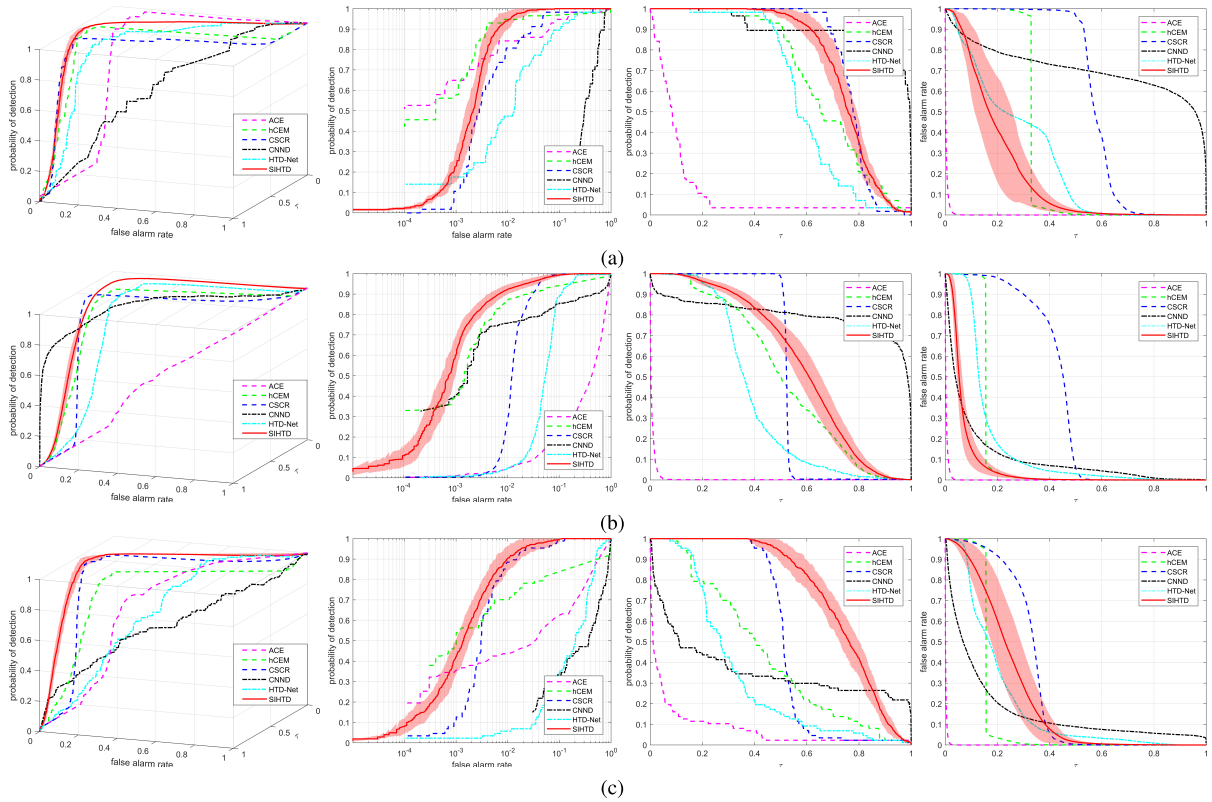


Fig. 8. Four types of ROC curves of the compared methods on three target data sets. From left to right: 3-D ROC curve, 2-D ROC curve of (P_F, P_D), 2-D ROC curve of (τ, P_D), and 2-D ROC curve of (τ, P_F). Note that the linewidth of SIHTD represents the standard deviation of the AUC value obtained in ten repeated tests. (a) AVIRIS-1. (b) AVIRIS-2. (c) AVIRIS-3.

TABLE II

ACCURACY AND COMPLEXITY (IN SECONDS) COMPARISON OF DIFFERENT APPROACHES ON THREE TARGET DATA SETS. BOLDFACE HIGHLIGHTS THE BEST RESULTS, WHILE UNDERLINE THE SECOND. THE RESULTS OF SIHTD ARE REPORTED IN THE FORM OF MEAN ± STANDARD DEVIATION

Method	AVIRIS-1						AVIRIS-2						AVIRIS-3					
	AUC_1	AUC_2	AUC_3	AUC_{oa}	Kappa	Time	AUC_1	AUC_2	AUC_3	AUC_{oa}	Kappa	Time	AUC_1	AUC_2	AUC_3	AUC_{oa}	Kappa	Time
ACE	0.9737	0.1196	**0.0054**	1.0879	**0.7313**	0.2050	0.5605	0.0090	**0.0054**	0.5641	0.0985	0.4126	0.8246	0.0780	**0.0036**	0.8990	0.4852	0.2150
hCEM	0.9726	0.6806	0.3300	1.3232	0.6327	**0.0927**	0.9573	0.5140	0.1611	1.3102	0.8604	**0.1684**	0.8662	0.4364	0.1597	1.1429	0.6552	**0.0923**
CSCR	0.9842	0.7758	0.5828	1.1772	0.5510	5.4404	0.9835	0.5244	0.4250	1.0829	0.7715	8.1709	0.9919	0.5226	0.3210	1.1935	0.6471	7.1414
CNND	0.6477	**0.9078**	0.7081	0.8474	0.0078	15.6900	0.8987	**0.8009**	0.1950	1.5046	0.8111	18.7776	0.6050	0.3617	0.1608	0.8059	0.0638	19.4187
HTD-Net	0.9613	0.6031	0.2652	1.2992	0.3355	17.7267	0.9335	0.3913	0.1671	1.1577	0.5388	18.0534	0.7524	0.3279	0.1993	0.8810	0.0532	17.4682
SIHTD	**0.9971** ±0.0004	0.7551 ±0.0302	0.1962 ±0.0544	**1.5560**	0.6664 ±0.0324	6.5392 ±0.2155	**0.9961** ±0.0012	0.5860 ±0.0332	0.0701 ±0.0138	**1.5120**	**0.9074** ±0.0134	3.4460 ±0.1668	**0.9969** ±0.0009	**0.7455** ±0.0316	0.2389 ±0.0531	**1.5035**	**0.7199** ±0.0131	4.7873 ±0.1355

CSCR on AVIRIS-2 is above that of SIHTD when $\tau < 0.53$ but shows otherwise when $\tau > 0.53$. In this case, quantitative AUC values can be applied for a more accurate comparison.

Table II shows four types of AUC and Kappa values of different approaches on three HSIs. The best results are boldfaced, while the second underlined. It can be seen that ACE always performs the best in background suppression but also the worst in target preservation. This point can be shown according to the visualization of detection results in Fig. 7. The CNND network can obtain a higher accuracy in target preservation than other methods in most cases, but it misclassified too many background pixels into target ones, which can be observed in the detection maps in Fig. 7. Our proposed SIHTD can achieve the highest AUC_1 and AUC_{oa} on three data sets, and usually, the second can achieve the highest AUC_2 and AUC_3. With respect to the overall detection accuracy AUC_{oa}, SIHTD can achieve an improvement of 23.28%, 0.74%, and 31.00% on AVIRIS-1, AVIRIS-2, and AVIRIS-3 in comparison with the second best results, respectively. For the kappa coefficient, SIHTD obtains the highest value on AVIRIS-2 and AVIRIS-3, and the second obtains the highest value on AVIRIS-1, which indicates that there is a higher coincidence between the detection map and the ground truth than other methods, that is, the proposed SIHTD can obtain better performance than most competitors when considering both true detection and false alarm. As for deep learning-based HTD methods, CNND cannot obtain a satisfactory performance since CNND measures the similarity/dissimilarity more

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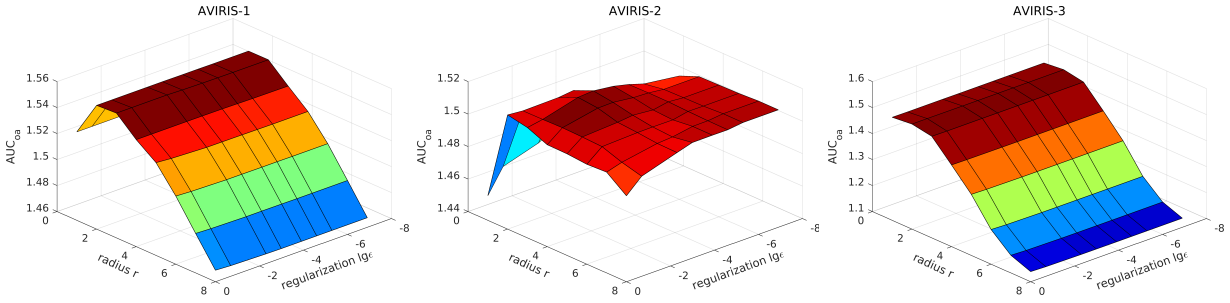


Fig. 9. Impact of parameters r and ϵ on detection accuracy of the proposed SIHTD method. From left to right: AVIRIS-1, AVIRIS-2, and AVIRIS-3. For clarity, we replace the ϵ axis with $\lg \epsilon$.

effectively between HSIs from the same spectrometer. Also, HTD-Net fails to detect targets, the possible reason is that the target signatures generated by U-autoencoder (U-AE) network have high similarity to the few prior-known target spectra and cannot introduce additionally useful information to training samples. The proposed method can successfully transfer similarity/dissimilarity measurement from source domain to target domain due to the novel semisupervised domain adversarial adaptation module and preserve the shape of target better due to the application of joint spatial–spectral feature and guided filter.

Moreover, complexity analysis of different methods is shown in Table II. The training time of three deep learning-based methods, CNND, HTD-Net, and our SIHTD, is around 400, 200, and 400 s on AVIRIS-1, AVIRIS-2, and AVIRIS-3, respectively. It is worth noting that the running time of HTD-Net does not include target and background sample generation and pixel-pair construction, which takes about 300 s in total. The computational complexity listed in Table II is only the testing time. Obviously, the nondeep-learning-based methods without training procedure consume much less time than deep-learning-based ones. The proposed SIHTD is the most efficient methods based on data-driven deep learning. Importantly, no repeated training is needed on the source domain repeatedly. Once the initial FSDL module is trained, only the target FSDL module needs to be updated through semisupervised domain adversarial adaptation on different target data. This can save a significant amount of time in practice.

D. Discussion

In this section, the impacts of several main parameters and efficiency of two major components in the proposed SIHTD are analyzed in detail.

1) Impact of the Guided Filter: The S^2TD module includes two main parameters, i.e., r and ϵ. The values of r and ϵ are in the sets $\{1, 2, \ldots, 8\}$ and $\{0.001, 0.005, 0.01, 0.02, 0.05, 0.1, 0.5, 1.0\}$, respectively. Their impacts on overall detection accuracy are verified on all data sets, as shown in Fig. 9. It can be observed from Fig. 9 that the sensitivity of SIHTD to parameters r and ϵ remains different on three HSIs. For the first and third data, SIHTD shows high sensitivity to r. This is because a too large r would lead to oversmooth of the detection map, thus causing accuracy decrease in detection of airplanes with small size. Fortunately, SIHTD tends to achieve relatively stable performance at a smaller r. Thus, it can be empirically adjusted in a locally small range. However, for the second data set, we can observe that SIHTD is also sensitive to the parameter ϵ, where the detection accuracy decreases dramatically if ϵ is too large. The reason is similar to the aforementioned situation, i.e., the greater ϵ, the larger degree of blur, and the worse detection. Therefore, parameters r and ϵ are suggested to set to a relatively small value in practice.

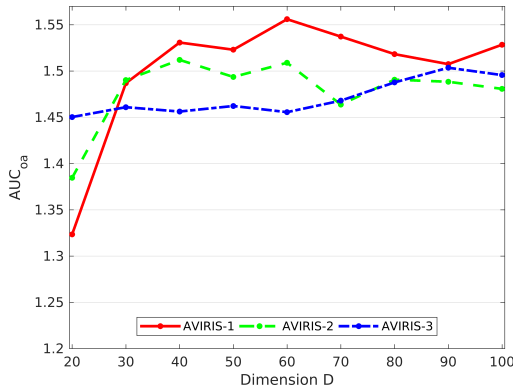


Fig. 10. Impact of the feature dimension D on detection accuracy of the proposed SIHTD method.

2) Impact of Feature Dimension: The third main parameter in SIHTD is the dimension D of the discriminative spatial–spectral feature learned by the SDAFL network. Fig. 10 shows the impact of D on overall detection accuracy when D is limited to the set $\{20, 30, 40, \ldots, 100\}$ on all data sets. First of all, it can be seen that with the increase of D, the AUC_{oa} value increases steadily as a whole, especially on data set AVIRIS-1. However, a too large D may not bring obvious breakthrough but slightly decrease detection accuracy. There are two possible reasons: first, too many features may impose redundancy and limit the ability of discriminative feature representation; and second, a large D would introduce more parameters in network training, increasing the risks of overfitting and computational complexity. For comprehensive consideration of accuracy and complexity, it is recommended that the dimension D is initialized to half of the band number.

3) Impact of the Patch Size: The fourth important parameter in our method is the spatial size w_p of the training samples.

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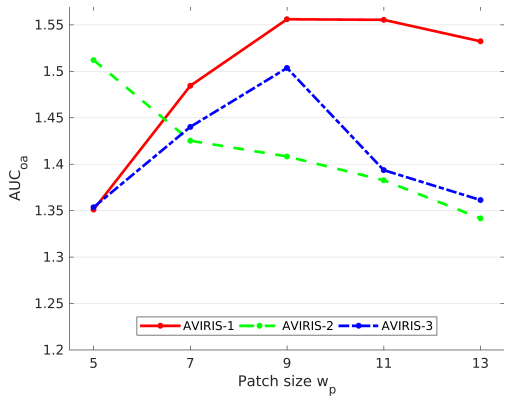


Fig. 11. Impact of the patch size w_p on detection accuracy of the proposed SIHTD method.

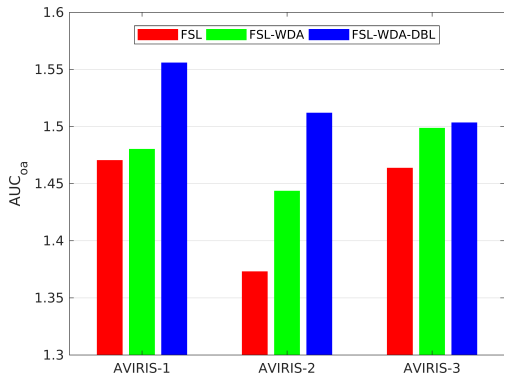


Fig. 12. Component analysis of the proposed SIHTD method on three target data sets.

The value of w_p is selected from the set $\{5, 7, 9, 11, 13\}$. Fig. 11 shows the overall detection accuracy on three HSIs when the value of w_p changes. According to Fig. 11, the best results of SIHTD can be obtained when $w_p = 9$, $w_p = 5$, and $w_p = 9$ for AVIRIS-1, AVIRIS-2, and AVIRIS-3, respectively. It is worth noting that the value of w_p is possibly related to the size of the detected objects, usually around the size of the smallest target by observing from Fig. 6. However, we can see that w_p for AVIRIS-2 is much smaller than the target size according to ground truth, and the detection accuracy decreases dramatically with the increase of w_p, which is inconsistent with our expectation. This is mainly because there exists probably a serious shadow of the target located in the middle area, which makes it quite different from other targets. The proposed method prefers to regard the bottom left part of it as target, and thus, the appropriate w_p is set to 5.

4) Component Analysis: To verify the effectiveness of each component of SIHTD, three versions are compared: 1) FSDL: FSDL introduced in Section II-B; 2) FSDL-WDA: WDA described in Section II-C1 is combined with FSDL; and 3) FSDL-WDA-DBL: DBL function described in Section II-C2 is further added to FSDL-WDA. Fig. 12 shows the overall detection accuracy of different methods on three data sets. The obvious improvements of FSDL-WDA over FSDL demonstrate that data alignment between source domain and target domain is quite important for our method. FSDL-WDA-DBL consistently obtains higher accuracy than FSDL-WDA, which indicates the effectiveness of our DBL function specifically designed for HTD.

IV. Conclusion

In this article, to alleviate the detection accuracy degradation caused by labeled samples insufficiency, class imbalance, and sensor-dependent transferability, we proposed a novel SIHTD method based on the SDAFL model for HTD task. Specifically, few-shot learning and domain adaptation are combined to bridge the gap between source domain with a large amount of labeled data and target domain with a very few labeled data, where attention mechanism and adaptive weighting strategy are additionally applied to avoid negative transfer. Furthermore, to ensure that the extracted features can provide positive influence on subsequent target detection, we introduced MDConv operation and an effective DBL function to strengthen the ability of deep feature representation. The experimental results demonstrate that the SIHTD achieves a better comprehensive performance than other competitors in HTD.

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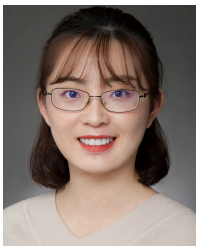
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Yanzi Shi (Graduate Student Member, IEEE) received the B.S. degree from the School of Telecommunications Engineering, Xidian University, Xi'an, China, in 2017. She is currently pursuing the Ph.D. degree with the State Key Laboratory of Integrated Service Networks, Xidian University.

Her research interests include hyperspectral target detection, feature extraction, deep learning, and urban remote sensing.



Jiaojiao Li (Member, IEEE) received the B.E. degree in computer science and technology, the M.S. degree in software engineering, and the Ph.D. degree in communication and information systems from Xidian University, Xi'an, China, in 2009, 2012, and 2016, respectively.

She was an exchange Ph.D. Student with Mississippi State University, Starkville, MS, USA. She is a Post-Doctoral Researcher and a Lecturer with Xidian University. Her research interests include hyperspectral remote sensing image analysis and processing, pattern recognition, and data compression.



Yunsong Li (Member, IEEE) received the M.S. degree in telecommunication and information systems and the Ph.D. degree in signal and information processing from Xidian University, Xi'an, China, in 1999 and 2002, respectively.

In 1999, he joined the School of Telecommunications Engineering, Xidian University, where he is a Professor. He is also the Director of the State Key Laboratory of Integrated Service Networks, Image Coding and Processing Center. His research interests focus on image and video processing, hyperspectral image processing, and high-performance computing.



Qian Du (Fellow, IEEE) received the Ph.D. degree in electrical engineering from the University of Maryland, Baltimore, MD, USA, in 2000.

She is the Bobby Shackouls Professor with the Department of Electrical and Computer Engineering, Mississippi State University, Starkville, MS, USA. She is also an Adjunct Professor with the College of Surveying and Geo-informatics, Tongji University, Shanghai, China. Her research interests include hyperspectral remote sensing image analysis and applications, pattern classification, data compression, and neural networks.

Dr. Du is a Fellow of the SPIE-International Society for Optics and Photonics. She received the 2010 Best Reviewer Award from the IEEE Geoscience and Remote Sensing Society. She was the Co-Chair of the Data Fusion Technical Committee of the IEEE Geoscience and Remote Sensing Society from 2009 to 2013 and the Chair of the Remote Sensing and Mapping Technical Committee of the International Association for Pattern Recognition from 2010 to 2014. She was the General Chair of the fourth IEEE GRSS Workshop on Hyperspectral Image and Signal Processing: Evolution in Remote Sensing held at Shanghai, China, in 2012. She has served as an Associate Editor for the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING, the *Journal of Applied Remote Sensing*, and the IEEE SIGNAL PROCESSING LETTERS. Since 2016, she has been the Editor-in-Chief of the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING.

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